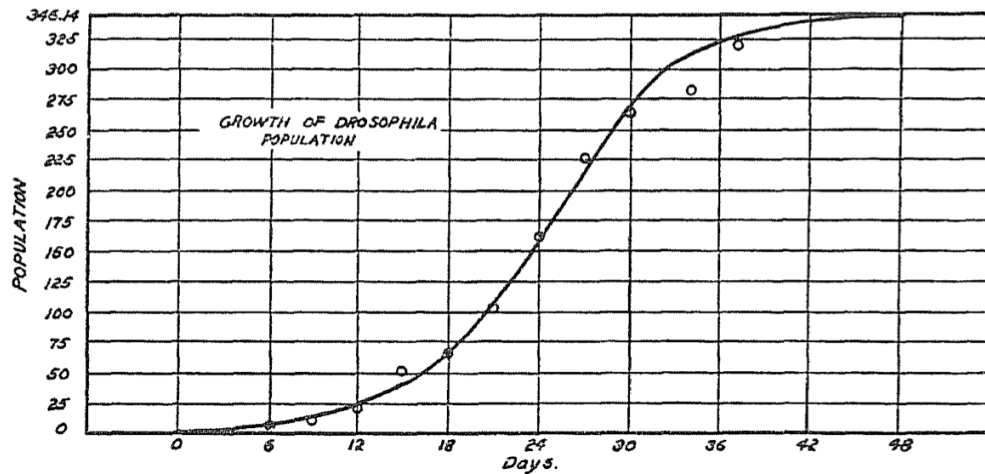


Lecture 12B Live Problems

Equilibrium for a single state variable. The logistic model.

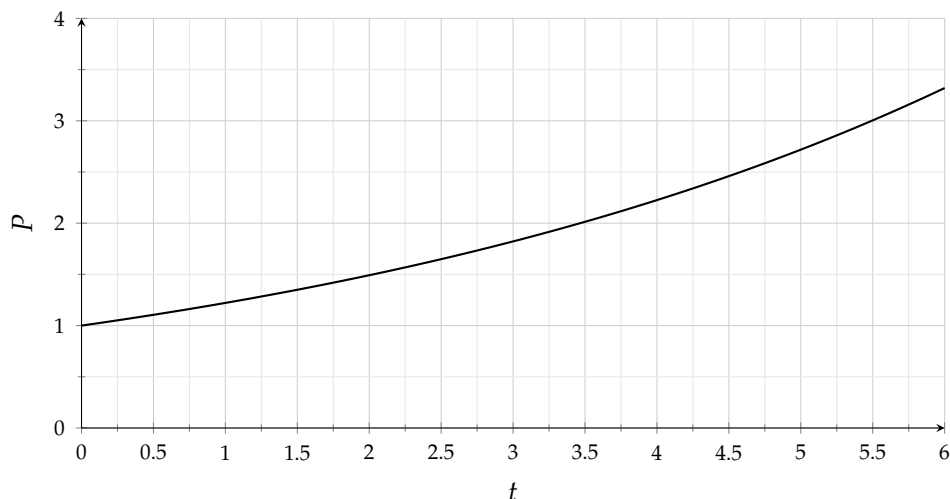
Points toward lab manual pages: **LO1, LO2, LO3, LO4** (section L3 S1)

1. Lotka's fruit flies. Here is a graph of a *Drosophila* population, from Lotka's book *Elements of Physical Biology* (1925).



- What is $P(24)$?
- What is $P'(24)$? (Also called dP/dt when $t = 24$.)

2. Exponential growth. Consider the change equation $P' = 0.2 P$. What does the time-series plot look like? What is the doubling time? What is the predictive equation?



3. The logistic change equation. $P' = 0.2 P - 0.0006 P^2$. Tabulate the rate of population change P' , with and without the crowding effect'' term, when $P = 10, 100, 200, 300, 400$.

P	P' without crowding term	P' with crowding term
10		
100		
200		
300		
400		

4. What are the equilibrium points for the change equation $P' = 0.2P - 0.0006P^2$?